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Magnetized white dwarfs with mixed poloidal-toroidal fields

Draft – a certified 2 $M_{\odot}$ configuration

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ABSTRACT

Context. A white dwarf well above the Chandrasekhar mass requires support beyond electron degeneracy. A strong internal magnetic field can supply it, but a purely toroidal field has no exterior structure and so no dipole for magnetic braking to act on.
Aims. We construct a 2 $M_{\odot}$ configuration carrying both components, and ask what a poloidal component costs when it is added to a toroidal-dominated equilibrium.
Methods. The toroidal field is solved self-consistently with the stellar structure by a self-consistent field iteration under a cold degenerate equation of state. A single solve carrying both components is not available in barotropy, where the toroidal field is confined to the closed-poloidal-line region and the reachable ratio is a few percent, so the poloidal component is solved on the converged density and imposed. Every configuration is gated on the virial error, on the star fitting inside its domain, and on the central density staying below the inverse-beta-decay threshold.
Results. The configuration reaches $M = 2.007\,M_{\odot}$ at $\rho_c = 10^9\,\mathrm{g\,cm^{-3}}$ with $E_{\mathrm{tor}}/|W| = 0.203$, a virial error of 9.9×10^{-5}, and a mass converging monotonically under mesh refinement. The field is strongly toroidal-dominated, $B_{\mathrm{t}}/B_{\mathrm{p}} = 884$ in energy, with a peak of 4.28×10^{13} G and an exterior dipole of 5.9×10^{10} G. The virial error of the pair tracks the imposed poloidal energy almost one for one, so a more balanced field leaves the certified regime: at $B_{\mathrm{t}}/B_{\mathrm{p}} = 8.8$ the virial error is twenty times the acceptance threshold.
Conclusions. A 2 $M_{\odot}$ white dwarf with a strong toroidal field and an exterior dipole is constructible and certifiable, at the cost of a poloidal component carrying a tenth of a percent of the magnetic energy. It is not shown to be stable: a field this toroidal is the configuration most exposed to the Tayler instability, which an axisymmetric calculation cannot test.

Key words. white dwarfs – stars: magnetic field – magnetohydrodynamics (MHD) – methods: numerical

1. Introduction

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2. Equation of state

The background stellar structure throughout this work is that of a fully degenerate, zero-temperature electron gas (ztwd; Chandrasekhar 1935). Pressure and density are both parametrized by the normalized Fermi momentum $x \equiv p_F/(m_e c)$,

$$P(x) = A\left[x(2x^2 - 3)\sqrt{x^2 + 1} + 3\sinh^{-1} x\right], \quad (1)$$

$$\rho(x) = B\,x^3, \quad (2)$$

where A is a fixed combination of fundamental constants (m_e, c, $\hbar$) and B depends on the composition only through the mean molecular weight per electron $\mu_e = 1/Y_e$, with Y_e the number of electrons per baryon. This is a purely mechanical equation of state: at fixed composition, pressure depends on density alone, with no temperature dependence. Consequently, once the configuration is restricted to be non-rotating and field-free, the entire stellar structure (mass, radius, density profile) is determined by exactly two physical parameters: the central density ρ_c and μ_e.

The specific enthalpy has a closed form,

$$H(x) = \frac{8A}{B}\left[\sqrt{1 + x^2} - 1\right], \quad (3)$$

which is what makes the self-consistent field (SCF) iteration of Sect. 3 tractable: at each iteration the density is recovered directly from the local enthalpy via the inverse relation $x(H) = \sqrt{(1 + HB/8A)^2 - 1}$, with $H \leq 0$ mapped to $\rho = 0$ (the stellar surface).

Near the surface ($x \ll 1$) the equation of state reduces to $\rho \propto H^{3/2}$, an effective polytropic index $n = 3/2$; in the relativistic core ($x \gg 1$) it approaches $\rho \propto H^3$, $n = 3$ – the marginal polytropic index at which the equilibrium mass becomes independent of central density, the origin of the Chandrasekhar mass limit.

Figure 1 plots Eqs. 1 and 2 and shows where those limits actually apply. The transition is slow: the adiabatic index $\Gamma = \mathrm{d}\ln P/\mathrm{d}\ln\rho$ takes about four decades in density to fall from 5/3 to 4/3, and neither limit is a good description through most of that range. It also places the configuration of Sect. 4. At its central density, $\rho_c = 10^9\,\mathrm{g\,cm^{-3}}$, the normalized Fermi momentum is $x = 8.0$ and $\Gamma = 1.343$ – within 0.7% of the relativistic limit. The core therefore sits essentially at the marginal index, which matters twice over here: at $n \simeq 3$ the equilibrium is only weakly sensitive to ρ_c and correspondingly delicate, and it is also the regime in which magnetic support translates most directly into mass, since the magnetic energy scales with radius the same way the gravitational and internal energies do.

At sufficiently high density, inverse beta decay (electron capture) changes the composition locally and the constant-μ_e cold equation of state above ceases to describe a real white dwarf.

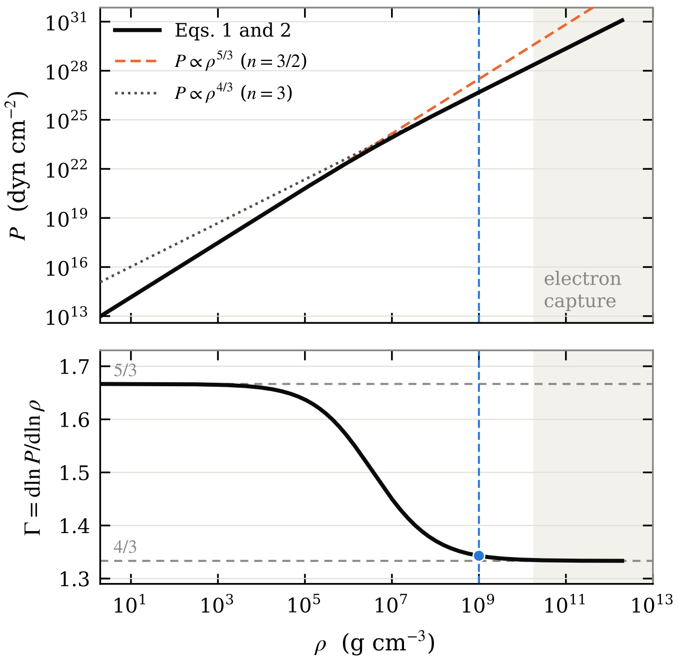

Fig. 1. The equation of state of Eqs. 1 and 2 for $\mu_e = 2$. Upper panel: $P(\rho)$ with the two polytropic limits, each normalized on the curve in the regime where it holds. Lower panel: the adiabatic index, which crosses from 5/3 to 4/3 over roughly four decades in density. The dashed vertical line marks the central density of the configuration of Sect. 4, and the point its adiabatic index there. The shaded region lies above the inverse-beta-decay threshold for $\mu_e = 2$, where this equation of state no longer describes a white dwarf.

We adopt the tabulated threshold densities of Boshkayev et al. (2013) for this validity limit; all central densities discussed below are kept below this threshold for the assumed composition.

3. Field equations

3.1. The toroidal branch

For a barotropic star the magnetic force must derive from a potential if hydrostatic balance is to close as a Bernoulli integral. For an axisymmetric, purely azimuthal field $\mathbf{B}_t = B_\varphi \hat{\mathbf{e}}_\varphi$ this restricts B_φ to

$$B_\varphi = K\rho^m \varpi^{2m-1}, \qquad m \geq 1, \tag{4}$$

with ϖ the cylindrical radius, K the amplitude and m the concentration towards the dense interior. The Lorentz force then follows from a potential, $\mathbf{f}_L/\rho = -\nabla M_{\rm tor}$ with

$$M_{\rm tor}(s) = \frac{mK^2}{4\pi(2m-1)}\, s^{2m-1}, \qquad s \equiv \rho\varpi^2, \tag{5}$$

which we verified symbolically for $m = 1, 3/2, 2, 3$. Hydrostatic equilibrium reduces to

$$H + \Phi + M_{\rm tor}(\rho\varpi^2) = C, \tag{6}$$

solved together with the Poisson equation for Φ and the equation of state of Sect. 2 by the self-consistent field iteration of Hachisu (1986). Because $M_{\rm tor}$ depends on ρ, the density update is no longer a direct inversion of the equation of state but a per-point root find; the left-hand side is monotonic in ρ, so the root is unique and safely bracketed.

Table 1. The adopted configuration. $\mu_e = 2$, $m = 1$, no rotation.

ρ_c (g cm^{-3})	1.0×10^9
M ($M_\odot$)	2.007
$R_{\rm eq}$ (km)	5207
$R_{\rm pol}$ (km)	5736
$R_{\rm pol}/R_{\rm eq}$	1.102
K	3.245×10^{-3}
$E_{\rm tor}/\lvert W\rvert$	0.2026
$B_{\varphi,\max}$ (G)	4.275×10^{13}
$B_{\rm pole}$ (G)	5.91×10^{10}
B_t/B_p (energy)	884
B_t/B_p (amplitude)	26.2
$E_{\rm pol}/E_{\rm mag}$	1.1×10^{-3}
VE	9.9×10^{-5}
$f_{\rm pol}$	0.142

3.2. The poloidal component, and why it is imposed

An axisymmetric poloidal field derives from a flux function $u(\varpi, z) = \varpi A_\varphi$,

$$B_\varpi = -\frac{1}{\varpi}\frac{\partial u}{\partial z}, \qquad B_z = \frac{1}{\varpi}\frac{\partial u}{\partial \varpi}, \tag{7}$$

with u obeying the Grad-Shafranov equation (Grad & Rubin 1958; Shafranov 1966) sourced by the current distribution, which we take linear in the flux.

A single self-consistent solve carrying both components is not available in this formulation, and the obstruction is not a numerical one. In barotropy the toroidal function must depend on the flux alone, $\beta = \beta(u)$; since there is no current outside the star, B_φ must vanish on every flux surface that reaches the surface, so the toroidal field is confined to the region of closed poloidal lines. The energy ratio reachable that way is a few percent, far from the toroidal-dominated configuration required below. We therefore solve the toroidal branch self-consistently for the structure, and impose the poloidal field on the converged density.

That imposition has a price, and Sect. 4.4 measures it: the imposed field's energy is precisely the term left unbalanced in the virial, so imposing a poloidal component carrying a fraction x of the binding energy costs a virial error of about x. The poloidal component adopted here is small enough that this cost stays below the acceptance threshold, which is what makes the configuration certifiable at all.

Purely toroidal and purely poloidal equilibria are each subject to a non-axisymmetric instability of azimuthal wavenumber $m = 1$ (Tayler 1973; Markey & Tayler 1973), which an axisymmetric formulation cannot test for. Sect. 5 returns to this.

4. Results

4.1. A certified 2 $M_\odot$ configuration

Table 1 reports the configuration this paper is about: a non-rotating white dwarf of 2.007 $M_\odot$ at $\rho_c = 10^9$ g cm^{-3}, carrying a toroidal field with $E_{\rm tor}/|W| = 0.2026$ and a poloidal component imposed on top.

The mass gain is the whole point of the toroidal field: the same central density and composition without a field give 1.347 $M_\odot$ at a radius of 2457 km, so the field is carrying +49% in mass and inflating the star by a factor 2.1 in equatorial radius. A 2 $M_\odot$ white dwarf of radius 5200 km is a large object by the standards of its mass – unsupported, a star this massive would be far more compact, and the size is the visible form of the support the field is providing. That gain is available at a central density well below the inverse-beta-decay threshold for $\mu_e = 2$ (1.94×10^{10} g cm^{-3}, Sect. 2), so the configuration is a white dwarf in the sense the equation of state can still describe.

The field is toroidal-dominated by a wide margin, $B_t/B_p = 884$ in energy and 26 in amplitude. The two differ because energy goes as the square and the components occupy different volumes: the toroidal field is confined to the interior while the poloidal one extends through the star and beyond it. Calling this a mixed field is accurate but should not be read as balanced – the poloidal component carries a tenth of a percent of the magnetic energy. What it does carry is the exterior structure, which the toroidal component cannot have at all: B_φ vanishes with the density, so a purely toroidal star has no field outside it and no dipole for magnetic braking to act on. The poloidal component supplies a surface dipole of 5.9×10^{10} G.

One number worth stating next to the field strength. The critical field at which Landau quantization changes the electron equation of state is $B_c = m_e^2c^3/e\hbar = 4.414\times10^{13}$ G, and the peak field here is $0.97\,B_c$ – just below it. The equation of state used has no field dependence, so a configuration above B_c would be inconsistent with its own microphysics; this one sits at the edge of that regime rather than inside it.

4.2. The field inside the star

Figures 2 and 3 show the field the previous section quantifies. Both take a form dictated by the axisymmetry: for an axisymmetric field $\mathbf{B}_p$ is perpendicular to ∇u, so the poloidal field lines *are* the contours of the flux function, exactly, and are drawn as such rather than integrated. The three-dimensional lines are integrated in cylindrical coordinates for the same reason – it keeps them on their flux surfaces by construction, and the drift of u along a traced line then measures the integration error rather than hiding it. Over the lines drawn it stays between 10^{-4} and 3×10^{-3}.

The meridional cut shows the star is prolate (Table 1), which is the deformation a toroidal field produces and the opposite of rotation's. B_φ is concentrated well inside the surface, vanishing on the axis where $\varpi \to 0$ and at the surface where the density does. The dashed contour marks $\rho = 10^{-3}\rho_c$ and sits far inside the stellar surface, which is the extended low-density envelope this configuration carries: the field lives in the dense interior, not in the envelope it inflates.

In three dimensions the same field is a set of nested tori wound by a tight helix. Following the drawn lines until each closes its meridional circuit gives between 17 and 20 turns around the axis per circuit. This is the twisted-torus geometry with the twist taken to an extreme: the poloidal circuit is barely traced out before the line has wrapped the axis a score of times.

The winding is a property of each flux surface and is not the amplitude ratio of Table 1 restated. That ratio is $\max|B_\varphi|/\max|\mathbf{B}_p|$, with the two maxima taken at different places in the star, so it is a global number rather than a local one. Along the lines drawn here the local $|B_\varphi|/|\mathbf{B}_p|$ averages between 35 and 255, varying by nearly an order of magnitude from surface to surface, while the winding number changes by less than a fifth. The two quantities answer different questions.

4.3. Convergence

Figure 4 shows the configuration under mesh refinement at fixed domain, halving Δr four times. The mass converges monotonically with shrinking increments – 0.15%, 0.04%, 0.01% – to $2.007\,M_\odot$, and $E_{tor}/|W|$ is converged to four significant figures at every resolution beyond the coarsest.

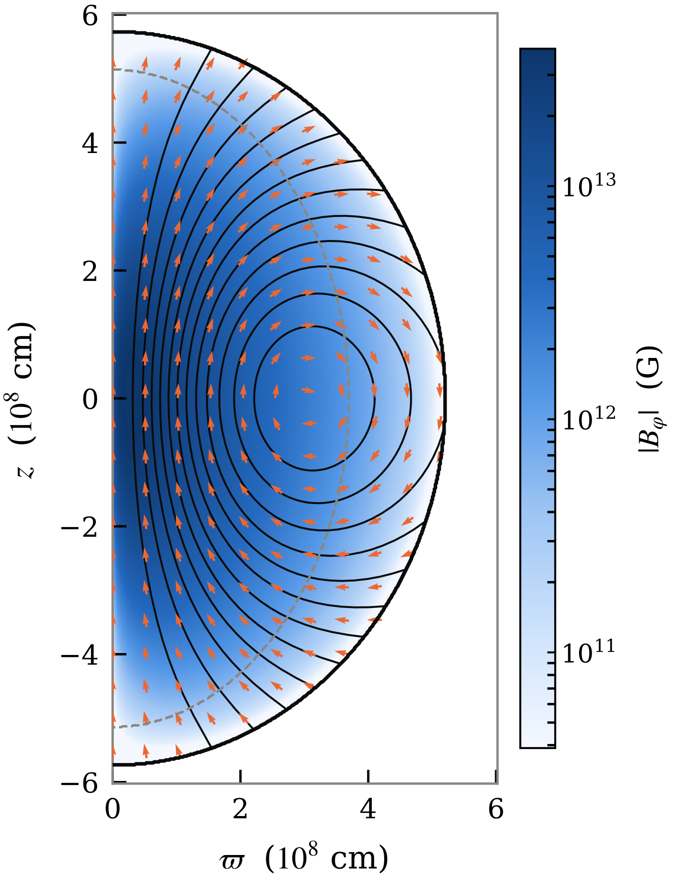

Fig. 2. Meridional cut of the adopted configuration. Colour is $|B_\varphi|$, out of the plane. Black contours are the poloidal field lines, which for an axisymmetric field are exactly the contours of the flux function; orange arrows give the poloidal direction, drawn as unit vectors because the poloidal amplitude spans orders of magnitude. The heavy black curve is the stellar surface, prolate as a toroidal field makes it; the dashed curve is $\rho = 10^{-3}\rho_c$.

The virial error stays below the 10^{-3} acceptance threshold at all four resolutions, but it is not monotonic: it dips at the second resolution and settles near 10^{-4}. We report this rather than smoothing it over, because it means the virial error at a single resolution is not an error estimate for this configuration; the mass convergence is. The same non-monotonicity appears at larger K in this branch, where it does cross the threshold; here it does not.

The domain was sized by the criterion that the star's polar radius clear the boundary, $f_{pol} = 0.142$ against the 0.2 limit adopted for this branch, and the configuration is reached by continuation in K from the field-free solution, since direct solves at this amplitude do not converge.

4.4. What the poloidal component costs

The poloidal field is imposed rather than solved for, so it leaves the pair out of virial balance by its own energy. Figure 5 measures that directly: as the poloidal amplitude is raised, the virial error of the combined configuration tracks $E_{pol}/|W|$ almost one for one, and the pair leaves the certified band once the exterior dipole exceeds roughly 10^{11} G.

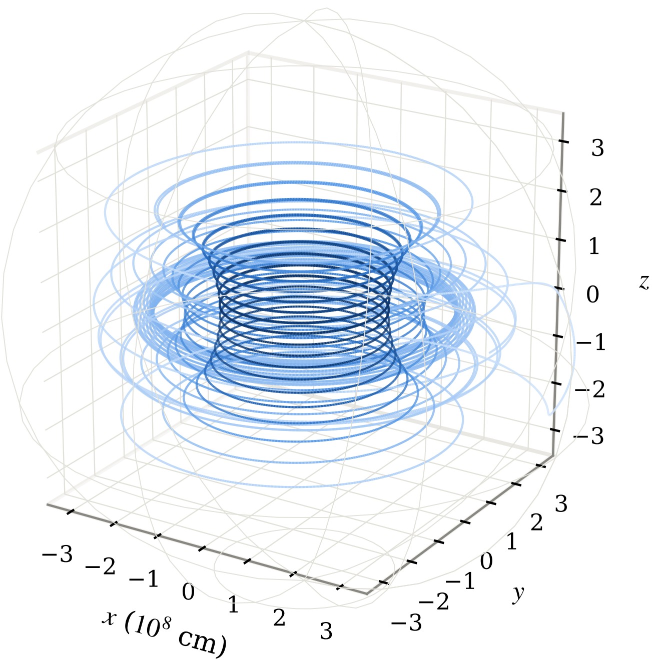

Fig. 3. Field lines of the same configuration in three dimensions, coloured by field strength, with the stellar surface as the faint wireframe. Lines lie on nested flux surfaces and wind 17 to 20 times around the axis for each circuit they complete in the meridional plane.

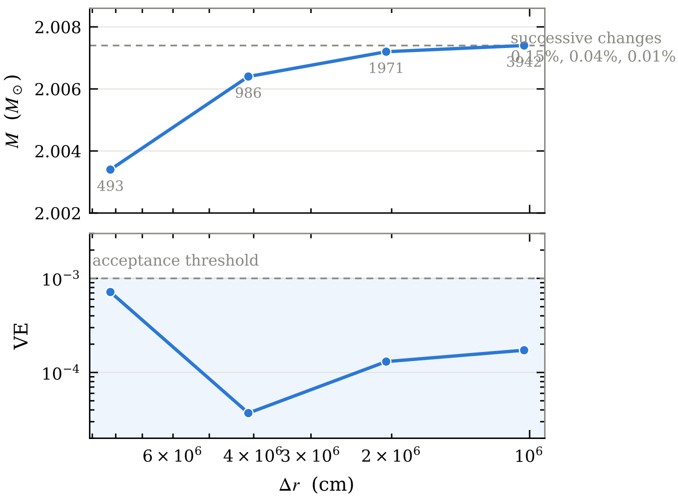

Fig. 4. Mesh convergence of the adopted configuration at fixed domain, labelled by radial resolution. Upper panel: the mass, converging monotonically to 2.007 $M_\odot$; the dashed line is the finest value. Lower panel: the virial error, which stays inside the acceptance band (shaded) at every resolution without being monotonic.

This sets the configuration adopted above. At $B_t/B_p = 884$ the virial error is 9.9×10^{-5}, an order of magnitude inside the threshold, and the exterior dipole is 5.9×10^{10} G. Reaching a more balanced field is possible but not certifiable by the standard applied here: at $B_t/B_p = 8.8$ the virial error is 2.3×10^{-2}, twenty times the threshold. A configuration in that range would have to be justified dynamically – loaded into an evolutionary calculation and allowed to relax – rather than presented as an equilibrium.

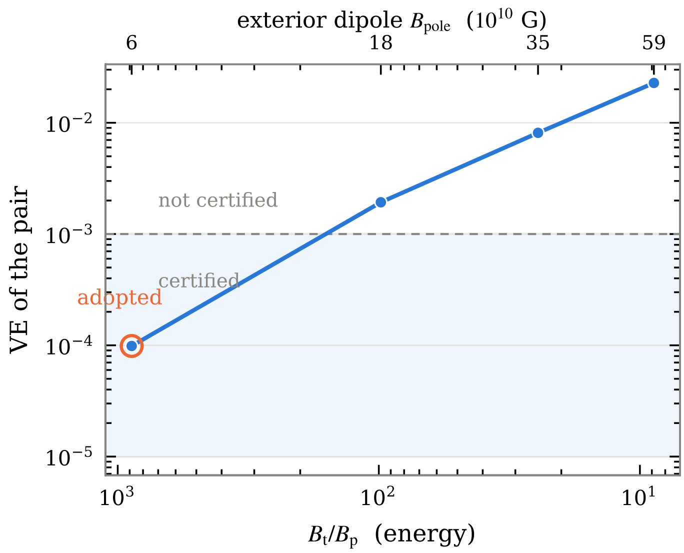

Fig. 5. The cost of the poloidal component. Moving left to right, the imposed poloidal amplitude rises, the field becomes less toroidal-dominated, and the exterior dipole (upper axis) grows; the virial error of the pair rises with it and leaves the certified band (shaded). The circled point is the configuration of Table 1.

5. Limitations

No stability test. This is a two-dimensional axisymmetric equilibrium, certified to satisfy hydrostatic and virial balance and nothing more. A field this strongly toroidal is precisely the configuration subject to the Tayler $m = 1$ instability (Tayler 1973; Markey & Tayler 1973), which is non-axisymmetric and therefore invisible to this formulation by construction. The poloidal component adopted here is far too weak to be the stabilizing agent that a mixed field is supposed to provide. Whether this configuration survives a non-axisymmetric perturbation is open, and answering it requires a three-dimensional magnetohydrodynamic evolution.

The poloidal component is not self-consistent. It is solved on the converged toroidal density and imposed, for the reason given in Sect. 3.2: barotropy does not deliver a strongly mixed equilibrium. The resulting virial error is small enough to certify, but the configuration is not a solution of a single set of equations carrying both components.

The equation of state does not respond to the field. At 0.97 B_c the peak field is at the edge of the regime where Landau quantization modifies the degenerate electron gas, and the inverse-beta-decay threshold used to bound ρ_c would itself shift with field (Das & Mukhopadhyay 2013). Neither effect is included. Staying below B_c keeps the configuration self-consistent in this respect, but only barely, and it forecloses raising K further at this central density.

Barotropy. The star is described by a cold, temperature-independent equation of state, so it has no entropy stratification and no buoyancy restoring force. Stable magnetic equilibria in stars are usually argued to rely on stable stratification, which this model does not have; whether that is a limitation of the model or a statement about degenerate stars is beyond what these calculations can settle.

Acknowledgements. TBD.

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